Regression Autopsy

Eight Ways Your Model Is Lying to You

Lecture Notes

Statistical and Machine Learning Lab

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These notes accompany the interactive Jupyter notebook series. All theorems stated here are validated by Monte Carlo simulation in the corresponding notebooks.

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1 Introduction

Regression is the most widely used tool in quantitative data analysis. It produces point estimates, standard errors, confidence intervals, p-values, and R^2 statistics—all of which look authoritative. In most observational analyses, at least one of these numbers is wrong.

These lecture notes formalise the mathematical foundations behind the eight core failure modes of ordinary least squares (OLS) regression. Each section corresponds to one interactive notebook in the accompanying lab series. The notes are designed to be self-contained: every claim is stated as a formal result and, in the lab, confirmed by Monte Carlo simulation.

1.1 The NovaMart Scenario

Throughout the course, we work with a single fictional company—**NovaMart**, an online retailer. The variables of interest are:

- Y: monthly revenue (thousands of dollars),
- X_1: advertising spend (thousands of dollars),
- X_2: number of sales staff,
- X_3: customer satisfaction score,
- U: unobserved market demand (hidden from the analyst).

Using a single recurring dataset creates continuity: when a coefficient changes from one notebook to the next, the student remembers the old value.

2 The Data Generating Process

Definition 2.1 (NovaMart DGP)**.** The data are generated according to the following structural equations:

$$U \sim \mathcal{N}(0,1), \tag{1}$$
$$X_1 = \gamma_1 U + \eta_1, \quad \eta_1 \sim \mathcal{N}(0,\sigma_\eta^2), \tag{2}$$
$$X_2 = \gamma_2 U + \eta_2, \quad \eta_2 \sim \mathcal{N}(0,\sigma_\eta^2), \tag{3}$$
$$\varepsilon \sim \mathcal{N}(0,\ h \cdot |X_1|)\,, \tag{4}$$
$$Y = \beta_0 + \beta_1 \log(X_1) + \beta_2 X_2 + \beta_U U + \varepsilon, \tag{5}$$
$$X_3 = \alpha\, Y + \eta_3, \quad \eta_3 \sim \mathcal{N}(0,\sigma_\eta^2). \tag{6}$$

The default parameter values are:

Parameter	Symbol	Default
Intercept	β_0	50
Ad spend effect	β_1	8
Staff effect	β_2	3
Demand effect	β_U	2
Endogeneity strength	γ_1	20
Staff–demand loading	γ_2	5
Heteroscedasticity	h	0.5
Bad-control strength	α	0.1
Noise scale	σ_η	1

Remark 2.1 (Causal ordering). The generation order is critical: first U, then (X_1, X_2), then Y, then X_3. The variable X_3 is *post-treatment*: it is caused by Y, not a cause of Y. It never appears in the true DGP for Y. Including X_3 as a control variable introduces collider bias (see Section 8).

2.1 Pathologies Embedded in the DGP

The single DGP simultaneously contains all of the following pathologies:

1. **Omitted variable bias:** U drives both X_1 and Y. Omitting U biases $\hat{\beta}_1$ upward.
2. **Heteroscedasticity:** $\mathrm{Var}(\varepsilon \mid X_1) = h \cdot |X_1|$; larger campaigns have more variable returns.
3. **Inflated significance:** with large n, trivially small effects become "significant."
4. **Misspecification:** the true relationship is $\log(X_1)$, not X_1.
5. **Overfitting:** adding polynomial terms inflates training R^2 but destroys test R^2.
6. **Collider bias / bad control:** X_3 is caused by Y; conditioning on it opens a spurious path.

Each notebook isolates and examines one pathology while the others remain present in the background.

3 Why Your Coefficient Is Wrong: Omitted Variable Bias

3.1 Setup

Suppose the true model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X_1, X_2] = 0, \tag{7}$$

but the analyst estimates the *short* regression

$$Y = \tilde{\beta}_0 + \tilde{\beta}_1 X_1 + \tilde{\varepsilon}. \tag{8}$$

Theorem 3.1 (Omitted Variable Bias Formula)**.** *Let $\hat{\beta}_1^{short}$ denote the OLS estimator of $\tilde{\beta}_1$ from the short regression (8), and let β_1 be the true coefficient in (7). Define the* auxiliary slope

$$\delta_1 \;=\; \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)}.$$

Then

$$\mathbb{E}[\hat{\beta}_1^{\text{short}}] \;=\; \beta_1 + \beta_2\, \delta_1. \tag{9}$$

The term $\beta_2 \delta_1$ is the ***omitted variable bias****.*

Proof. Write the OLS estimator in population form. From the short regression,

$$\text{plim}\ \hat{\beta}_1^{\text{short}} = \frac{\mathrm{Cov}(X_1, Y)}{\mathrm{Var}(X_1)} = \frac{\mathrm{Cov}(X_1,\, \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon)}{\mathrm{Var}(X_1)}.$$

Since $\mathbb{E}[\varepsilon \mid X_1, X_2] = 0$ implies $\mathrm{Cov}(X_1, \varepsilon) = 0$, we obtain

$$\text{plim}\ \hat{\beta}_1^{\text{short}} = \beta_1 + \beta_2 \cdot \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)} = \beta_1 + \beta_2\, \delta_1. \qquad \square$$

Key Insight

The bias has a *sign* and a *direction*. In the NovaMart DGP with default parameters, $\beta_2 = 3$ and $\delta_1 > 0$ (because both X_1 and X_2 are driven by U), so the bias is *positive*: the short regression overestimates the effect of ad spend. The OVB formula predicts $\mathbb{E}[\hat{\beta}_1^{\text{short}}] \approx 2.3 + 2.0 \times 1.2 = 4.7$, which is confirmed by Monte Carlo simulation (mean ≈ 4.698 over 5,000 replications).

3.2 When OVB Cannot Be Signed

The formula (9) generalises to multiple omitted variables. If k variables are omitted, the bias on $\hat{\beta}_1$ is a weighted sum of terms $\beta_j \delta_{1j}$, where each δ_{1j} is the partial regression coefficient of X_j on X_1 after partialling out included regressors. In general, the sign of the total bias is indeterminate without knowledge of all omitted variables.

3.3 The Proxy Variable Approach

The Repair

If the true confounder U is unobservable, one can include a *proxy* $\tilde{U}$ that is correlated with U. If $\tilde{U}$ explains a fraction $q \in [0, 1]$ of the variation in U, the residual bias is approximately $(1 - q)$ times the original bias. At $q = 1$ (perfect proxy), the bias vanishes; at $q = 0$, the full bias remains.

Warning

Including X_3 (customer satisfaction) as a control makes things *worse*, not better. Because X_3 is caused by Y, conditioning on X_3 opens a collider path and introduces additional bias.

4 Why Your Uncertainty Is Wrong: Heteroscedasticity

4.1 The Classical Variance Formula

Under the standard linear model $Y = X\beta + \varepsilon$ with $\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \sigma^2 I_n$, the variance of the OLS estimator is

$$\mathrm{Var}(\hat{\beta}_{\mathrm{OLS}} \mid X) = \sigma^2 (X^\top X)^{-1}. \tag{10}$$

Definition 4.1 (Heteroscedasticity)**.** The error term is *heteroscedastic* if $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma_i^2$ with σ_i^2 not constant across i. In the NovaMart DGP, $\sigma_i^2 = h \cdot |X_{1,i}|$: larger ad campaigns have more unpredictable returns.

Theorem 4.1 (Variance Under Heteroscedasticity)**.** *When $\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \Omega \neq \sigma^2 I_n$, the OLS estimator $\hat{\beta}_{OLS}$ remains* unbiased *but its variance becomes*

$$\mathrm{Var}(\hat{\beta}_{OLS} \mid X) = (X^\top X)^{-1} X^\top \Omega\, X\, (X^\top X)^{-1}. \tag{11}$$

*This is the **sandwich formula**. The classical formula (10) underestimates or overestimates the true variance, depending on the correlation structure between X and the error variances.*

4.2 Consequences for Inference

If the classical standard errors are used under heteroscedasticity:

- Confidence intervals have incorrect coverage. In the NovaMart simulation, the nominal 95% CI achieves only ≈82% coverage.
- t-statistics and p-values are unreliable.
- The coefficient estimates themselves are *not* biased—only the uncertainty is wrong.

4.3 The Breusch–Pagan Test

Definition 4.2 (Breusch–Pagan Test)**.** Regress the squared OLS residuals $\hat{\varepsilon}_i^2$ on the regressors X. Under H_0: homoscedasticity, the test statistic nR^2 from this auxiliary regression follows χ_p^2 asymptotically, where p is the number of regressors (excluding the constant).

4.4 Robust Standard Errors

The Repair

The **White–Huber–Eicker** ("sandwich") standard errors are computed as

$$\widehat{\mathrm{Var}}(\hat{\beta}_{\mathrm{OLS}}) = (X^\top X)^{-1} \left(\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \, x_i x_i^\top \right) (X^\top X)^{-1},$$

where $\hat{\varepsilon}_i$ are the OLS residuals. These standard errors are consistent under heteroscedasticity of unknown form. With robust SEs, the 95% CI recovers ≈95% coverage.

Warning

Robust standard errors perform poorly in small samples. When $n = 20$, the sandwich estimator is more biased than the classical estimator. The crossover point depends on the severity of heteroscedasticity, but as a rough guide, $n \geq 50$ is typically needed for reliable sandwich SEs.

5 Why Your Significance Is Wrong: Effect Size and Power

5.1 The Anatomy of a t-Statistic

The t-statistic for testing $H_0 : \beta_j = 0$ is

$$t = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)}. \tag{12}$$

Since $\mathrm{SE}(\hat{\beta}_j) = O(1/\sqrt{n})$, for any fixed $\beta_j \neq 0$:

$$|t| \;\to\; \infty \quad \text{as } n \to \infty.$$

Key Insight

With enough data, *every* non-zero effect becomes statistically significant, no matter how small. With $n = 50{,}000$ observations (NovaMart at store-month level), customer satisfaction has $p < 0.001$ despite a coefficient of \$0.003 per satisfaction point—economically meaningless.

5.2 Statistical Power

Definition 5.1 (Power)**.** The **power** of a test is the probability of correctly rejecting H_0 when H_0 is false:

$$\text{Power} = P(|t| > c_\alpha \mid \beta_j \neq 0)\,,$$

where c_α is the critical value at significance level α.

Proposition 5.1 (Power Increases with n and Effect Size)**.** *For a two-sided t-test at level α with true coefficient β_j and standard deviation σ,*

$$\textit{Power} \approx \Phi\left(\frac{|\beta_j|}{\sigma/\sqrt{n}} - z_{\alpha/2} \right),$$

where Φ is the standard normal CDF and $z_{\alpha/2}$ is the upper $\alpha/2$ quantile. Power increases with both $|\beta_j|$ (effect size) and $\sqrt{n}$.

5.3 The Multiple Testing Problem

Running m independent hypothesis tests at level α produces

$$\mathbb{E}[\text{false positives}] = m \cdot \alpha.$$

With $m = 1{,}000{,}000$ tests at $\alpha = 0.05$, we expect 50,000 false positives.

The Repair

The **Bonferroni correction** tests each hypothesis at level α/m, controlling the family-wise error rate at α. For genome-wide association studies ($m \approx 10^6$), the corrected threshold is $\alpha/m = 5 \times 10^{-8}$. Always report effect sizes and confidence intervals alongside p-values.

6 Why Your Model Is Wrong: Specification Error

6.1 The Problem of Functional Form

Suppose the true relationship is $Y = f(X) + \varepsilon$ for some nonlinear function f, but we fit the linear model $Y = X\beta + \varepsilon$.

Theorem 6.1 (Linear Projection Under Misspecification)**.** *If $Y = f(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$, and we estimate $Y = X\beta + u$ by OLS, then the OLS estimator converges to*

$$\beta^* = \arg\min_{\beta} \mathbb{E}\left[(Y - X\beta)^2\right],$$

which is the best linear approximation to $f(X)$ in the mean-squared-error sense. This β^* equals the true β only if f is linear.

In the NovaMart DGP, $f(X_1) = \beta_1 \log(X_1)$. A linear model in X_1 provides a reasonable approximation within the observed data range but diverges catastrophically when extrapolating.

6.2 The Extrapolation Danger

If the true relationship is $Y = 8\log(X_1) + \cdots$ and we fit $Y = \tilde{\beta}_1 X_1 + \cdots$:

- At $X_1 = 100$ (within sample): moderate overestimation.
- At $X_1 = 2{,}000$ (extrapolation): linear prediction overshoots by a factor of ≈3.
- The confidence band widens but does *not* contain the truth.

The Repair

Fit the correctly specified model by transforming the predictor:

$$Y = \beta_0 + \beta_1 \log(X_1) + \beta_2 X_2 + \varepsilon.$$

Extrapolation accuracy improves dramatically. However, even the log model is a local approximation—at extreme extrapolation, all parametric models eventually fail.

7 Why Your R^2 Is Lying: Overfitting

7.1 The Bias–Variance Decomposition

Theorem 7.1 (Bias–Variance Decomposition)**.** *For any estimator $\hat{f}$ of the regression function f, the expected prediction error at a point x_0 decomposes as*

$$\mathbb{E}\Big[(Y - \hat{f}(x_0))^2\Big] = \underbrace{\Big(f(x_0) - \mathbb{E}[\hat{f}(x_0)]\Big)^2}_{\textit{Bias}^2} + \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\textit{Variance}} + \underbrace{\sigma_\varepsilon^2}_{\textit{Irreducible noise}} . \tag{13}$$

Proof. Let $\mu = \mathbb{E}[\hat{f}(x_0)]$. Then

$$\begin{aligned}\mathbb{E}[(Y - \hat{f})^2] &= \mathbb{E}[(f + \varepsilon - \hat{f})^2] \\ &= \mathbb{E}[(f - \hat{f})^2] + \mathbb{E}[\varepsilon^2] + 2\mathbb{E}[\varepsilon(f - \hat{f})] \\ &= \mathbb{E}[(f - \hat{f})^2] + \sigma_\varepsilon^2,\end{aligned}$$

where the cross-term vanishes because ε is independent of $\hat{f}$ (trained on separate data) and has mean zero. Now expand:

$$\mathbb{E}[(f - \hat{f})^2] = (f - \mu)^2 + \mathbb{E}[(\hat{f} - \mu)^2] = \mathrm{Bias}^2(\hat{f}) + \mathrm{Var}(\hat{f}). \qquad \square$$

7.2 Overfitting in Practice

Adding polynomial terms to a regression always increases training R^2 (since R^2 is monotonically non-decreasing in the number of regressors). In the NovaMart demo:

Degree	**Training R^2**	**Test R^2**
1	0.71	0.69
3	0.82	0.78
7	0.91	0.63
15	0.97	< 0

A degree-15 polynomial achieves $R^2 = 0.97$ on training data but predicts *worse than the sample mean* on held-out data.

7.3 Cross-Validation

Definition 7.1 (k-Fold Cross-Validation)**.** Partition the data into k equally sized folds. For each fold j, train the model on all data except fold j and compute the prediction error on fold j. The CV estimate of prediction error is

$$\mathrm{CV}(k) = \frac{1}{k}\sum_{j=1}^{k} \mathrm{MSE}_j \, .$$

The Repair

Using cross-validation, the log-linear model (which matches the true DGP) achieves the lowest CV error, outperforming all polynomial models. The right model family matters more than the degree of flexibility.

Warning

Cross-validation itself can overfit if the number of candidate model specifications searched is large. The best CV R^2 inflates with the breadth of the search.

8 Why Your Causal Claim Is Wrong: Endogeneity and Bad Controls

8.1 Endogeneity

Definition 8.1 (Endogeneity)**.** A regressor X_j is *endogenous* if $\mathrm{Cov}(X_j, \varepsilon) \neq 0$. This can arise from omitted variables, measurement error, or simultaneous causation (feedback loops).

In the NovaMart DGP, revenue and ad spend are mutually related through demand U. OLS captures the correlation but cannot distinguish causal direction.

8.2 Collider Bias from Bad Controls

The causal graph for NovaMart is:

$$U \longrightarrow X_1 \longrightarrow Y \longrightarrow X_3, \qquad U \longrightarrow Y.$$

Including X_3 (satisfaction) as a regressor conditions on a descendant of Y, which opens a spurious association path through Y.

Proposition 8.1 (Collider Bias)**.** *If $X_3 = \alpha Y + \eta_3$ (i.e., X_3 is a post-treatment variable), then the regression of Y on (X_1, X_3) yields a biased estimate of the causal effect of X_1 on Y, even if X_1 is otherwise exogenous. The bias arises because conditioning on a descendant of Y partially conditions on Y itself.*

Warning

The advice "control for everything available" is wrong. Controlling for post-treatment variables or colliders *introduces* bias rather than removing it. Variable selection for causal inference requires understanding the causal structure, not just statistical association.

8.3 Instrumental Variables

Definition 8.2 (Instrument)**.** A variable Z is a valid *instrument* for X in the equation $Y = X\beta + \varepsilon$ if:

1. **Relevance:** $\mathrm{Cov}(Z, X) \neq 0$.
2. **Exclusion:** $\mathrm{Cov}(Z, \varepsilon) = 0$ (the instrument affects Y only through X).

Theorem 8.1 (IV Consistency)**.** *If Z is a valid instrument, the IV estimator*

$$\hat{\beta}_{IV} = (Z^\top X)^{-1} Z^\top Y \tag{14}$$

is consistent for β:

$$\operatorname{plim} \hat{\beta}_{IV} = \operatorname{plim} \frac{Z^\top Y/n}{Z^\top X/n} = \frac{\mathrm{Cov}(Z, Y)}{\mathrm{Cov}(Z, X)} = \beta.$$

In the NovaMart context, if ad prices varied randomly across regions (a natural experiment), the regional price variation serves as an instrument for ad spend.

Warning

Weak instruments ($\mathrm{Cov}(Z, X) \approx 0$) cause the IV estimator to be highly variable and unreliable. When the first-stage F-statistic is below 10, IV confidence intervals are extremely wide and the estimator may be more biased than OLS.

9 The Real World: When All Pathologies Strike at Once

9.1 The Lalonde Dataset

Notebook 7 transitions from simulated data to the Lalonde (1986) dataset, a benchmark in causal inference. The question: does a job training programme increase earnings?

- **Observational estimate (OLS):** negative coefficient—training appears to *reduce* earnings.
- **Experimental estimate (RCT):** positive and large (≈$1,800).
- The two estimates do not even have the same sign.

All standard diagnostics pass: VIFs are low, Breusch–Pagan is non-significant, robust SEs barely change. The problem is not in the model—it is in the *data*. The observational and experimental groups differ systematically on unobservables.

9.2 Sensitivity Analysis: Cinelli–Hazlett Framework

Definition 9.1 (Robustness Value)**.** The **robustness value** (RV) is the minimum strength of association (measured in partial R^2) that an unobserved confounder would need to have with both the treatment and the outcome to explain away the observed result.

Formally, define

$$\mathrm{RV}_{q,\alpha} = \min\{r : \text{adjusted estimate crosses } q \cdot \hat{\beta} \text{ at level } \alpha\}.$$

A small RV indicates fragility: a weak confounder suffices to nullify the result. A large RV indicates robustness.

Key Insight

The Lalonde observational estimate has a very small robustness value: the result is fragile. A confounder with even modest partial R^2 with both treatment and outcome would explain away the finding. This is why the observational and experimental estimates disagree.

10 The Redemption: Regression Discontinuity Design

10.1 When Design Earns Trust

Regression works brilliantly when the world provides a natural experiment. The **regression discontinuity design** (RDD) exploits a sharp threshold to identify causal effects.

Definition 10.1 (Sharp RDD)**.** Suppose treatment is assigned deterministically by a *running variable* R at a cutoff c:

$$D_i = \mathbf{1}(R_i \geq c).$$

Under continuity of potential outcomes at c, the **local average treatment effect** at the cutoff is

$$\tau_{\mathrm{RDD}} = \lim_{r \downarrow c} \mathbb{E}[Y \mid R = r] - \lim_{r \uparrow c} \mathbb{E}[Y \mid R = r]. \tag{15}$$

Theorem 10.1 (RDD Identification)**.** *If $\mathbb{E}[Y_0 \mid R = r]$ and $\mathbb{E}[Y_1 \mid R = r]$ are continuous at $r = c$, then*

$$\tau_{RDD} = \mathbb{E}[Y_1 - Y_0 \mid R = c].$$

The design eliminates confounding locally *at the cutoff because units just above and just below the threshold are nearly identical in all characteristics.*

10.2 Example: Scholarship Cutoff

Students with test scores just above a threshold receive scholarships; those just below do not. Near the cutoff, the two groups are essentially identical. The jump in outcomes at the threshold identifies the causal effect of the scholarship.

10.3 Bandwidth Choice

The RDD estimator faces a bias–variance tradeoff controlled by the *bandwidth* h:

- **Narrow bandwidth:** low bias (observations are close to the cutoff), high variance (few observations).
- **Wide bandwidth:** low variance (more observations), high bias (observations far from the cutoff may not be comparable).

Optimal bandwidth selection (e.g., Imbens–Kalyanaraman or Calonico–Cattaneo–Titiunik) balances these two forces.

The Repair

The RDD does not require controlling for confounders. The design itself ensures comparability near the cutoff. When the autopsy report is run on the RDD estimate, all indicators are green: the regression is trustworthy because the *design* is trustworthy.

11 What "Controlling For" Actually Means: The Frisch–Waugh–Lovell Theorem

11.1 Motivation

When we say "the coefficient on X_1, controlling for X_2," we do *not* mean "holding X_2 constant." We mean: the coefficient from regressing the part of Y unexplained by X_2 on the part of X_1 unexplained by X_2.

Theorem 11.1 (Frisch–Waugh–Lovell)**.** *Consider the regression $Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$. Let $M_2 = I - X_2(X_2^\top X_2)^{-1}X_2^\top$ be the residual-maker matrix that projects orthogonally to the column space of X_2. Then*

$$\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y. \tag{16}$$

That is, $\hat{\beta}_1$ from the full regression is identical *(to numerical precision) to the coefficient obtained by:*

1. *Regressing Y on X_2 and taking residuals: $\tilde{Y} = M_2 Y$.*
2. *Regressing X_1 on X_2 and taking residuals: $\tilde{X}_1 = M_2 X_1$.*
3. *Regressing $\tilde{Y}$ on $\tilde{X}_1$ (no intercept).*

Proof. The OLS normal equations for the partitioned model $Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$ give

$$\begin{pmatrix} X_1^\top X_1 & X_1^\top X_2 \\ X_2^\top X_1 & X_2^\top X_2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} X_1^\top Y \\ X_2^\top Y \end{pmatrix}.$$

From the second block, $\hat{\beta}_2 = (X_2^\top X_2)^{-1} X_2^\top (Y - X_1\hat{\beta}_1)$. Substituting into the first block and simplifying:

$$\begin{aligned} X_1^\top X_1 \hat{\beta}_1 + X_1^\top X_2 (X_2^\top X_2)^{-1} X_2^\top (Y - X_1\hat{\beta}_1) &= X_1^\top Y \\ X_1^\top [I - X_2(X_2^\top X_2)^{-1} X_2^\top] X_1 \hat{\beta}_1 &= X_1^\top [I - X_2(X_2^\top X_2)^{-1} X_2^\top] Y \\ X_1^\top M_2 X_1 \hat{\beta}_1 &= X_1^\top M_2 Y. \end{aligned}$$

Hence $\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y$. □

Key Insight

"Controlling for" means residualising—stripping out the linear relationship with the control variables from *both* sides of the equation. The partial regression plot (also called an added-variable plot) visualises this: it plots $\tilde{Y}$ against $\tilde{X}_1$. The slope of this scatterplot *is* $\hat{\beta}_1$ from the full multiple regression.

12 Diagnostic Toolkit Summary

The following diagnostics are used across all notebooks:

Diagnostic	Tests For	Interpretation
Residuals vs. Fitted	Nonlinearity, heteroscedasticity	Residuals should show no pattern; funnel shape indicates heteroscedasticity
Normal Q–Q plot	Non-normality of residuals	Points should lie on the 45° line
Scale–Location	Heteroscedasticity	$\sqrt{\lvert\text{standardised residuals}\rvert}$ should be flat against fitted values
Residuals vs. Leverage	Influential observations	Points with high leverage *and* large residuals (Cook's $d > 1$) are influential
VIF	Multicollinearity	$\text{VIF}_j > 10$ suggests problematic collinearity
Breusch–Pagan	Heteroscedasticity	H_0: homoscedasticity; reject if $p < \alpha$
Durbin–Watson	Autocorrelation	Statistic near 2 indicates no autocorrelation
Jarque–Bera	Non-normality	H_0: residuals are normally distributed

Warning

Passing all diagnostics does not guarantee a valid causal estimate. In the Lalonde example (Notebook 7), every diagnostic passes, yet the OLS estimate has the wrong sign. Diagnostics check the model's internal consistency, not whether the research design is sound.

13 Theorem Map: How the Results Connect

The theorems in this course form a logical chain where each generalises the previous:

NB	Theorem	Generalisation Relationship
1	Omitted Variable Bias	Foundation: bias from omitting a relevant variable
2	Sandwich Variance	Relaxes $\mathrm{Var}(\varepsilon) = \sigma^2 I$; OVB is a special case of endogeneity
3	Power Analysis	$t = \hat{\beta}/\,\mathrm{SE}(\hat{\beta})$; builds on correct SEs from NB 2
4	Linear Projection	Best linear approx. to $f(X)$; relaxes f linear
5	Bias–Variance Tradeoff	$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Var} + \sigma^2$; generalises specification error
6	IV Consistency	If $\mathbb{E}[\varepsilon X] \neq 0$ but $\mathbb{E}[\varepsilon Z] = 0$; generalises OVB repair
7	Sensitivity Analysis	Robustness value quantifies fragility of any estimate
8	RDD Identification	Continuity at cutoff $\Rightarrow$ causal identification
1.5	Frisch–Waugh–Lovell	What "controlling for" means algebraically

The meta-lesson: each theorem relaxes an assumption that justified the previous conclusion. Classical variance is a special case of the sandwich estimator. The linear model is a special case of all models. OVB is a special case of endogeneity.

14 Conclusion

These notes have formalised the eight core failure modes of OLS regression demonstrated in the Regression Autopsy lab series:

1. Omitted variable bias shifts your coefficient.
2. Heteroscedasticity corrupts your standard errors.
3. Large samples make trivial effects "significant."
4. Misspecification makes extrapolation fiction.
5. Overfitting makes memorisation look like learning.
6. Bad controls and endogeneity destroy causal claims.
7. Real data confronts you with all failures simultaneously.
8. Sound research design (RDD) earns regression its trust back.

The unifying lesson is not that regression should be abandoned, but that its outputs must be accompanied by an honest assessment of the assumptions under which they are valid. The diagnostics tell you whether the *model* is internally consistent; only the *research design* can tell you whether the estimate is credible.

This doesn't mean you should stop using regression.
It means you're finally ready to start.